

E - Unbalanced ABC Substrings

分值: 450450450 分

Problem Statement

You are given a string SSS of length NNN consisting of the three characters A, B, and C. There are $N(N+1)2^{\frac{N(N+1)}{2}}2N(N+1)$ non-empty substrings of SSS ; find how many of them satisfy the following condition:

给定长度为 NNN 的字符串 SSS , 仅包含 A、B、C 三种字符。

SSS 共有 $N(N+1)2^{\frac{N(N+1)}{2}}2N(N+1)$ 个非空子串, 请计算其中满足以下条件的子串数量:

- The numbers of occurrences of A, B, and C are all distinct from each other.
- A、B、C 三者的出现次数互不相同。

Count two substrings separately if they are taken from different positions in SSS , even if they are identical as strings.

即使两个子串的内容完全相同, 只要它们在 SSS 中的位置不同, 就视为不同的子串, 需分别计数。

What is a substring?

什么是子串?

A **substring** of SSS is a string obtained by deleting zero or more characters from the beginning and zero or more characters from the end of SSS . For example, AB is a substring of ABC, but AC is not a substring of ABC.

SSS 的**子串**是指通过删除 SSS 开头的零个或多个字符、以及结尾的零个或多个字符得到的字符串。例如, AB 是 ABC 的子串, 但 AC 不是 ABC 的子串。

Constraints

- $1 \leq N \leq 2 \times 10^5$
- $|S| = N$

- NNN is an integer.
- SSS is a string consisting of the three characters A, B, and C.
- $1 \leq N \leq 2 \times 10^5$
- $|S| = N$
- NNN 是整数。
- SSS 是仅由 A、B、C 三种字符构成的字符串。

Input

The input is given from Standard Input in the following format:

输入按照以下格式从标准输入给出：

NNN
 SSS

Output

Output the answer on a single line.

将答案输出在一行中。

Sample Input 1

6
AABBCC

Sample Output 1

4

Deleting 000 characters from the beginning and 333 characters from the end of SSS gives AAB, which satisfies the condition.

Deleting 111 character from the beginning and 222 characters from the end of SSS gives ABB, which satisfies the condition.

Deleting 222 characters from the beginning and 111 character from the end of *SSS* gives BBC, which satisfies the condition.

Deleting 333 characters from the beginning and 000 characters from the end of *SSS* gives BCC, which satisfies the condition.

No other substrings satisfy the condition.

从 *SSS* 开头删除 000 个字符、结尾删除 333 个字符，得到满足条件的 AAB。从 *SSS* 开头删除 111 个字符、结尾删除 222 个字符，得到满足条件的 ABB。从 *SSS* 开头删除 222 个字符、结尾删除 111 个字符，得到满足条件的 BBC。从 *SSS* 开头删除 333 个字符、结尾删除 000 个字符，得到满足条件的 BCC。没有其他满足条件的子串。

Sample Input 2

6
ABCABC

Sample Output 2

0

No substring satisfies the condition.

不存在满足条件的子串。

Sample Input 3

10
ACABCAABAB

Sample Output 3

17